CURRICULUM VITAE

ANDREA DEL GAUDIO

OFFICE ADDRESS:
iMMC – Institute of Mechanics, Materials and Civil Engineering
Université Catholique de Louvain, Louvain-la-Neuve, BE
Phone: +39 (333) 8954-327
e-mail: delgaudio@uiowa.edu; andrea.delgaudio@uclouvain.be

PROFESSIONAL PREPARATION:

University of Naples "Federico II", Napoli, Italy: Civil & Env. Engrg., Phd Student., 2021-2024
University of Naples "Federico II", Napoli, Italy: Civil & Env. Engrg., M.S., 2021
University of Naples "Federico II", Napoli, Italy: Civil & Env. Engrg., Bachelor, 2019

APPOINTMENTS:

2026-present	FSR-Incoming Postdoctoral Associate, Institute of Mechanics Materials and Civil Engineering, iMMC, UCLouvain, Louvain-la-Neuve, Belgium
2025-2026	Postdoctoral Associate, Department of Civil and Environmental Engineering, University of Naples "Federico II", Napoli, Italy (1.5 years)
2024-may/july	Visiting Ph.d. Student, IGB – Leibniz-Institut für Gewässerökologie und Binnenfischerei, Berlin, Germany (3 months)
2023-2024	Visiting Ph.d. Student, University of Iowa & IIHR – Hydroscience and Engineering, Iowa city, USA (8 months)

HONORS:

-2026- **FSR/FNRS Incoming Postdoctoral Fellowship promoted by UCLouvain.** This funding aims to create favorable conditions for transnational mobility of highly qualified researchers through the award of postdoctoral fellowships.
Success rate ~10%.
-2026- **MSCA seal of excellence** for the project ME-DAM: "A novel Modelling and Ecological impact framework for assessing DAM-break induced flood hazard, flood mitigation and consequences of dam reconstruction". Score: 90+
-2022- Univ. Naples "Federico II" Department of Civil Architectural and Environmental Engineering **(DICEA) Faculty Excellence Award for *cursus studiorum in M.S. of Environmental Engineering***
-2021- **"Buon Compleanno Federico II"** University Excellence Award for university *cursus studiorum*, University of Naples "Federico II".

CURRENT PRINCIPAL FIELDS OF INTEREST:

My main areas of research are:

1) **Flood propagation and mitigation of geological hazards associated with floods and dam break events:** development of numerical tools and experimental analysis for flood propagation and dam break flows of Newtonian and non-Newtonian fluids with a focus on mud flows using fully-3D, non-hydrostatic large eddy simulation (LES) models with deformable free surface capabilities.
2) **Eco-hydraulics and Shallow flows**: investigation of flow hydrodynamics in vegetated channels using large eddy simulations with a focus on characterizing flow structure inside and downstream of vegetation patches and drag forces acting on the plant stems as part of an US NSF-funded project.
3) **Morpho-dynamics**: characterization of active channels in ephemeral rivers using DinSar satellite data. I developed a new procedure for determining sediment transport and morphodynamical changes after floods events.
4) **Pump intake flows:** develop models and perform state-of-the-art eddy resolving numerical simulations to predict flow and vortical structures in pump bays.

Development of numerical tools to assess design of complex multi-bay pump stations and effectiveness of vortex suppressing devices.

OTHER PROJECTS:

1) **Mass oscillations in pressurized pipes**: The Laplace transform is used to solve a system of differential equations derived from the mass and energy conservation laws. The phenomenon of mass oscillations occurs in pressurized pipelines between a supply tank at a constant elevation and a piezometric well when the flow rate is changed by a valve located downstream. The study focused on characterizing the temporal behavior of the free surface oscillations of the piezometric well.

2) **Analytical derivation of the solution to the wave equation with homogeneous initial conditions:** this problem can be interpreted as the behavior of a flow disturbance inside a pressurized pipeline triggered by the closure of a valve located downstream. The analysis assumes uniform initial flow conditions and negligible head losses. The pipeline draws water from a reservoir while maintaining a constant water level.

3) **Implementation of a high-order (space and time) finite-volume (FV) solver for Shallow Water Equations:** The code I develop within a Matlab environment simulates 2D shallow water dynamics on a predefined domain using the MUSCL scheme for higher-order accuracy and the HLL-Rusanov numerical flux to compute fluxes at cell interfaces. It models friction effects based on the local Reynolds numbers for the flow. The time-stepping algorithm integrates the solution based on dynamically adjusted time steps to ensure numerical stability. The code includes visualization of the solution and compares it to external data, outputting results for further analysis.

4) **Workshop on Advanced Measurements Techniques and Experimental Research:** Conducted a series of hydraulic flume scale experiments at Técnico de Lisboa and analyzed the resulting experimental data with Matlab.

5) **Fluvial Bio-geomorphology Summer School:** Multidisciplinary training experience integrating geomorphology, hydraulics, and vegetation dynamics. I acquired fundamental insights into the basic bio-geomorphology processes at play in river systems and learned how to use and develop tools to understand how physical and biological processes interact to shape the river's ecosystem. I acquired cutting-edge knowledge in fluvial bio-geomorphology, riparian vegetation modelling, and the use of remote sensing technology.

REVIEWER:

Scientific Journals: Mechanica, Springer Nature (Nature Portfolio), Journal of Mountain Science, Physics of Fluids, Journal of Fluid Mehcnics, Journal of Hydraulic Engineering, Journal of Hydraulic Research.

SCIENTIFIC MEETINGS:

- Chair of the Special Session "Mini symposium on dense fluid flows interacting with the environment" for the 11th IAHR International Symposium on Environmental Hydraulics (ISEH2027) (June 1-4, 2027) - University of Iowa, USA.
- River Flow 13th International Conference on Fluvial Hydraulics (July 1-4, 2026) - University of Thessaloniki, GR.
- 41st International Association for Hydro-Environment Engineering and Research (IAHR) World Congress (June 22-27 2025) – National University of Singapore, SG.
- 6th International Symposium on Shallow Flows (September 15-18, 2025) "Effect of porous rectangular patch of emerged vegetation on turbulent coherent structures in channel flow" Turin, Italy.
- River Flow 12th International Conference on Fluvial Hydraulics (September 2-6, 2024) - University of Liverpool, UK.
- American Geophysical Union Annual Meeting *AGU23 December 11-15, 2023*, San Francisco, CA, USA.
- 5th EWaS International Conference (12-15 July 2022) - University of Naples Federico II (Via Partenope 36, 80121 Naples IT)

TEACHING EXPERIENCE:

ICAR02 Hydraulics infrastructures, A.A. 2025-2026, Degree Course in Environmental Engineering (Bachelor), Università degli Studi di Napoli Federico II, Prof. Eng. Francesco De Paola.

ICAR02 Hydraulics infrastructures, A.A. 2024-2025, Degree Course in Environmental Engineering (Bachelor), Università degli Studi di Napoli Federico II, Prof. Eng. Francesco De Paola.

ICAR02 Hydraulics infrastructures, A.A. 2023-2024, Degree Course in Environmental Engineering (Bachelor), Università degli Studi di Napoli Federico II, Prof. Eng. Francesco De Paola.

ICAR02 Hydraulics infrastructures, A.A. 2022-2023, Degree Course in Environmental Engineering (Bachelor), Università degli Studi di Napoli Federico II, Prof. Eng. Francesco De Paola.

Tutor for Geometry and Calculus, A.A. 2023-2024 , Degree Course in Civil Engineering (Bachelor), Università degli Studi di Napoli Federico II, Prof. Brunetti.

LIST OF PUBLICATIONS

<u>Journal Papers</u>

Published or in press:

1- Del Gaudio, Andrea; La Forgia, Giovanni; Constantinescu, George; Di Cristo, Cristiana; De Paola, Francesco; Iervolino, M.; Leopardi, Andrea; Vacca, Andrea. (2024), Modelling the impact of a dam–break wave on a vertical wall, *Earth Surface Processes and Landforms,* 10.1002/esp.5817

2- Del Gaudio, Andrea, Constantinescu, George; Di Cristo, Cristiana; De Paolo, Francisco; Vacca, Andrea (2024) "Large eddy simulation of power-law fluid dam break wave impacting against a vertical wall." *Physical Review Fluids* 9, 074801, doi/10.1103/PhysRevFluids.9.074801

3- Del Gaudio, Andrea, La Forgia, Giovanni., De Paola, Francesco, Di Cristo, Cristiana., Iervolino, M., Leopardi, Andrea., Vacca, Andrea (2024). Dam-Break waves over mobile bed. *Advances in Water Resources*, 193, 104801.

4- Del Gaudio, Andrea; Constantinescu, George; De Paola, Francesco; Di Cristo, Cristiana; Vacca, Andrea (2025), Turbulent dam-break waves of Newtonian and non-Newtonian fluids, *Journal of Fluid Mechanics,* doi:10.1017/jfm.2025.10636.

5- Del Gaudio, A., Constantinescu, G., De Paola, F., Zeng, J., & Ansar, M. (2026). Detached Eddy Simulation Model to Evaluate Design of Pump Stations. *Journal of Hydraulic Engineering*, *152*(3), 05026002.

6- Lama, G. F. C., Del Gaudio, A., Barman, J., Handique, A., De Paola, F., Chirico, G. B., & Termini, D. (2026). Influence of emergent and submerged plants on local hydrodynamics and turbulence anisotropy in vegetated rivers. *Journal of Hydrology*, 136182.

7- Del Gaudio, A., Constantinescu, G., & De Paola, F. (2026). Drag forces and flow structure around and inside a square array of emerged cylinders in an open channel. *Journal of Fluid Mechanics*, *1044*, A14.

Under Review:

8- Del Gaudio, A., Constantinescu, G., De Paola, F., Zeng, J., & Ansar, M. (2026). A numerical study of scale and turbulence model effects on the pump bay vortices and pump column swirl at a multi-bay pump station. *Journal of Hydraulic Engineering.*

9- Del Gaudio, A., Constantinescu, G., & De Paola, F. (2026). Turbulent lock exchange waves of Newtonian and non-Newtonian fluids propagating over inclined surfaces. *Physical Review Fluids.*

10- Wang, H., Figueroa, M., Del Gaudio, A., Ishi, K., Sanchez Huanca, J., Constantinescu, G. S., & Chamorro, L. P. (2026). Configurational control of drag and multiscale wake dynamics in patchy canopies. *Journal of Fluid Mechanics.*

11- Guadagno, V., Czapiga, M. J., Del Gaudio, A., Del Giudice, G., Di Cristo, C., Di Martire, D., Leopardi, A., Repola, L., & Viparelli, E. (2026). A physics-based channel extraction method from geospatial data based on structural connectivity. *Water Resources Research* (manuscript in preparation).

Conference Proceedings

1-Del Gaudio, A.; De Paola, F.; Di Cristo, C.; La Forgia, G.; Leopardi, A.; Vacca, A. (2022) "Experimental Investigation and Numerical Evaluation of the Free Surface of a Dam Break Wave in the Presence of an Obstacle." Environ. Sci. Proc. 2022 https://doi.org/10.3390/ environsciproc2022021023
2-A del Gaudio, G Constantinescu, C Di Cristo, F De Paola, A Vacca (2023) "Turbulent Dam-Break waves of non-Newtonian fluids" AGU23
3-A. Del Gaudio, G. La Forgia, G. Constantinescu, F. De Paola, C. Di Cristo, M. Iervolino, A. Leopardi, A. Vacca. (2024) "Laboratory Experiments and Numerical simulations of dam-break waves impacting a vertical wall" 12th River Flow Conference on Fluvial Hydraulics, Liverpool, UK 2024